

Capture protocol (follow exactly)

Before any capture (2 min). Switch on every light. Open curtains and blinds. **Open every interior door fully:** an open door can be measured and links the rooms; a closed one cannot. Move anything standing in a doorway. Phone charged $\geq 30\%$ with ≥ 3 GB free.

Tier	Phone	App	Time
A · LiDAR	iPhone 12 Pro or newer <i>Pro</i> , or iPad Pro (has LiDAR)	Stray Scanner (free, App Store, by Stray Robots)	~1 min per room
B · Video	any iPhone 15 or newer	built-in Camera app	~1 min per room
C · Photos	any iPhone 15 or newer	built-in Camera app	4–8 photos per room

A · LiDAR: Stray Scanner, one continuous recording

1. Install **Stray Scanner** from the App Store; open it and allow camera access. Do not change its settings.
2. Stand in the entrance. Tap **record**. Hold the phone **upright at chest height**, about **1 m from the walls**.
3. Walk **slowly (half your normal pace)** along the walls of the first room. Every 1–2 steps, **tilt up until the wall–ceiling line shows, then down until the wall–floor line shows**.
4. At each doorway: stop **about 1.5 m before it** and hold the **whole door frame** in view for 2 seconds; walk through slowly, then **turn round and show the frame for 2 seconds from the other side**.
5. Visit every room the same way. **Finish where you started**, pointing at the first wall for 2 seconds. Tap **stop**.
6. **Hand-off:** Files app → On My iPhone → Stray Scanner → press and hold the newest folder → **Compress** → AirDrop / USB / cloud the .zip to the computer.

B · Video: Camera app, one continuous clip

1. Settings → Camera → Record Video: **1080p at 30 fps**. Turn **off** Action mode and Cinematic. Use the **1×** lens; **never zoom**.
2. Hold the phone **sideways (landscape)** at chest height. Start recording at the entrance and walk the **same path as A**: slow, along the walls, tilting up to the ceiling line and down to the floor line every 2 steps, every room, back to the start. Doorways exactly as A step 4 (2 s from 1.5 m, both sides).
3. *Optional, improves accuracy:* lay **one sheet of A4 or Letter paper flat on the floor** of the first room and film it from standing height for 2 seconds.
4. **Hand-off:** AirDrop to a Mac, or USB to Windows with Settings → Photos → Transfer to Mac or PC → **Keep Originals**. **Do not send through WhatsApp or other messengers** (they strip the data the pipeline needs).

C · Photos: Camera app, one folder per room

1. Settings → Privacy & Security → Location Services → **Camera** → **While Using** (the compass stored in each photo helps join rooms). Photo mode, **1×** lens, **no zoom, no Portrait, flash off**. Hold the phone **sideways**.
2. **Walls:** stand with your **back against the middle of a wall** and photograph the **opposite wall**, so that **both of its corners, the floor line and the ceiling line** are in the picture. Repeat from the middle of every wall: a rectangular room gives 4 photos.
3. **Doorways:** for each doorway, stand 1 m inside the room facing it, so the **whole door frame and part of the next room** are in the picture: 1 photo per doorway.
4. Hallways: one photo from each end looking along it, plus one per doorway. **At most 8 photos per room**.
5. *Optional, improves accuracy:* lay **one sheet of A4 or Letter paper flat on the floor**, uncovered and away from rugs, where at least two photos of the room show it; move it to the next room with you.
6. **Hand-off:** make **one folder per room**, named after the room (01 Living, 02 Hall, ...), put that room's original photos in it (HEIC is fine), and put all room folders inside one folder. Same messenger warning as B.

Avoid (all tiers)

Running or fast turns (take at least 2 s per quarter turn) · pointing at a mirror or window for more than a second · lights off · fingers over the camera · walking backwards · recordings over 8 minutes (split the property by floor).

Run (one command per capture)

groundplan run <the .zip, video file, or photos folder>, then open runs/<name>/plan.png and runs/<name>/report.html.